

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459039.8743 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.067
Transit depth (Rp/Rs)^2	2.19 +/- 1.99 [%]
Orbital Inclination (inc)	85.04 +/- 1.73
Airmass coefficient 1 (a1)	1.05 +/- 0.62
Airmass coefficient 2 (a2)	0.05 +/- 0.86
Scatter in the residuals of the lightcurve fit is	66.35 %
Best Comparison Star	None
Optimal Aperture	0.87
Optimal Annulus	3.75
Transit Duration (day)	0.032 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.067 vs 0.1714 (deviation 0.28σ)
Duration fit vs published	0.032 d vs 0.0892 d (deviation 1.46σ)
Mid-time O–C	-5.3 min
Depth SNR	1.7
Residual scatter	66.35 %

Light curve

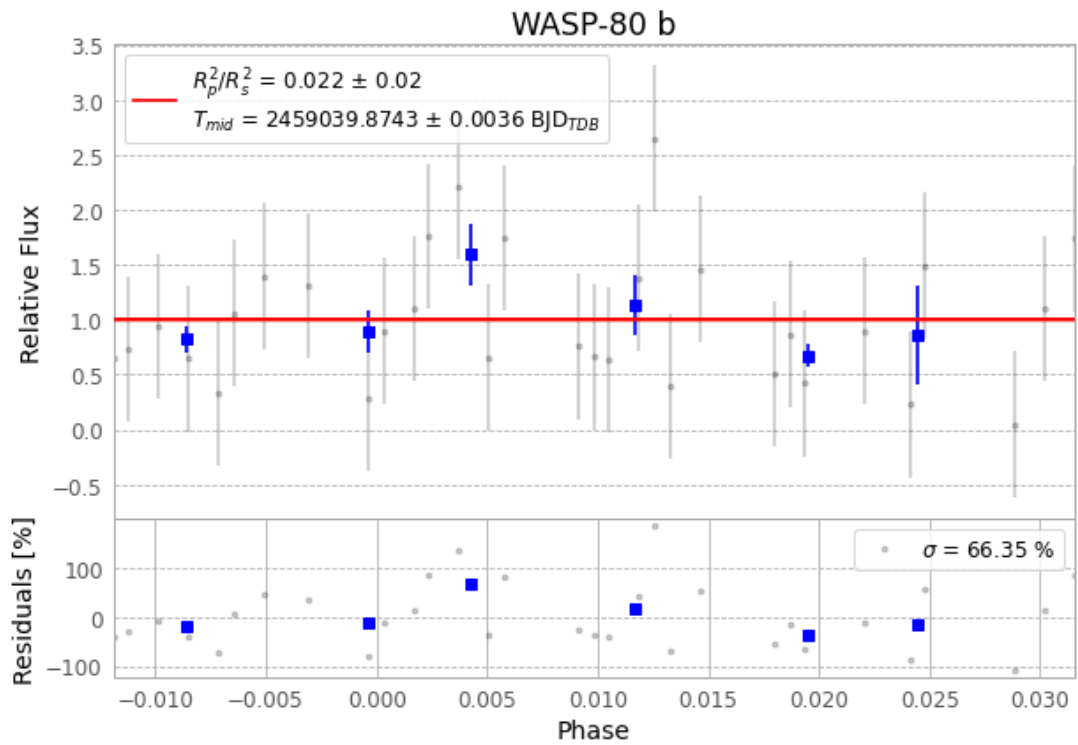


Figure 1 — FinalLightCurve_WASP-80 b_2020-07-09.png (SHA-256 0ef32b6c30e8afd8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [511, 379], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6876e8f96cfb3840...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

65 FITS frames (43 MB), WASP-80200709075712.FITS → WASP-80200709110912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-80-200709.log (5f38524cf6b5...), FinalLightCurve_WASP-80 b_2020-07-09.png (0ef32b6c30e8...), FinalLightCurve_WASP-80 b_2020-07-09.csv (3d9d3e192eb7...)

Compute resources

Wall time 112.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 23:47Z.